1.3.13 Water Proof Finish

Water proof finish gives the resistance of water to the fabric. To increase the resistance various substance like paraffin, acid, resin, tannin, drying oils, alum or alumina salt carbonate magnesia are applied on to the fabric. The number of times the coating is done varies depending upon the substance used. A water-proof fabric is completely moisture proofed. The fabric is coated or laminated with a film of natural or synthetic rubber or plastic, such as vinyl or polyurethane to give proofing effect. Water proof finishes adversely affect the comfort property of the fabric as they limit the passage of air and possesses a rather firm and a bad hand feel.

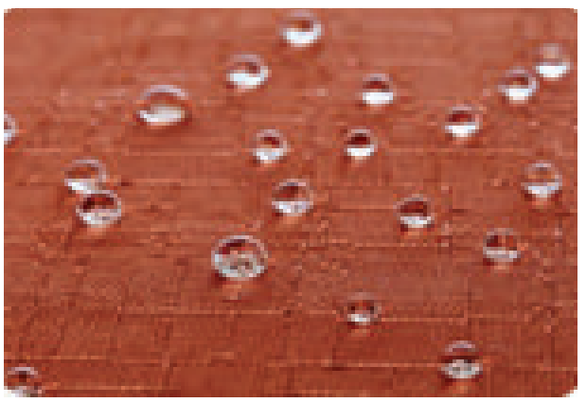


Figure 1.13 Water Proof Fabric

1.3.14 Water-Repellent Finish

Water repellent finishes are chemical finishes which resist the penetration of water into the fabric but permits the passage of moisture or air. When the fabric becomes very wet the water eventually passes through it. The yarns are coated with the repellent material like wax which permits the passage of air and vapour between the interlacing in the fabric. Water and the other liquid remain on the surface in small bead rather than spreading out and getting absorbed. The chemicals used are silicones, fluorocarbons, paraffin's etc. Some chemicals used for water repellency are also stain repellent. The combination of fabric finish and structure is important to achieve water repellent finish because it depends on the surface tension and fabric penetrability. The combination can make fabric which is stain resistance, having soft feel and a good drape.

Water repellent finish can be of both durable and non-durable types. The non-durable repellents are easily removed in laundering or dry cleaning. They do not provide satisfactory resistance to oily liquids. Durable repellent finish can be either repellent to water or oil or both. Fluro-carbon compounds have excellent durability to both dry cleaning and laundering.

DO YOU KNOW?

Do you know the oldest finished fabric?

Fulling cloth is one of the oldest finished fabrics. It is seen in Scots hat woven fabric, done centuries ago using engraving to change.



1.3.15 Flame Retardant Finish

Flame retardant finishes is one of the varieties of functional finishes. They play an important role on textiles by providing safety and giving escape time from a potential hazard. When a fire starts flame,

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